

# ASSIGNMENT 1

## Steady-State Heat Conduction

Maximum marks: 100

### Instructions

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1. Derive the finite-difference (or control-volume) discretization from first principles before writing any code.
2. Wherever an analytical solution is given, validate your numerical solution against it at several interior points and report the maximum absolute and percentage errors, not just a visual comparison of plots.
3. Solve linear system  $[A]\{T\} = \{b\}$  by (i) a *direct* method (Gaussian elimination / LU decomposition, exploiting bandedness where possible) and (ii) *iterative* methods (Jacobi, Gauss–Seidel, and SOR, as specified in each problem). Report the number of iterations, wall-clock time, and memory footprint as the grid is refined, and discuss which approach you would recommend and why. (Problem 3 instead uses a Picard/Newton iteration around a tridiagonal solve, and Problem 4 may be solved by any method consistent with its own tasks.)
4. Perform a grid-refinement (convergence) study for every problem: at least three grid sizes, with error plotted or tabulated against  $\Delta x$ ,  $\Delta r$ , or  $\Delta\theta$  as appropriate.
5. State all convergence criteria (e.g.  $\max|T^{p+1} - T^p| < 10^{-6}$ ), initial guesses, and relaxation factors used.
6. Submit your source code (or a link to a repository) as an appendix, along with all plots and tables requested.

| Group 1<br>Problems: 1, 8 | Group 2<br>Problems: 2, 7 | Group 3<br>Problems: 3, 6 | Group 4<br>Problems: 4, 5 |
|---------------------------|---------------------------|---------------------------|---------------------------|
| 230885                    | 230135                    | 230518                    | 231088                    |
| 230905                    | 230153                    | 230640                    | 231096                    |
| 251020623                 | 251050046                 | 251290008                 | 252050605                 |
| 261050013                 | 261050051                 | 261050080                 | 261050611                 |
| 261050022                 | 261050054                 | 261050086                 | 261050612                 |
| 261050028                 | 261050058                 | 261050089                 | 261180013                 |
| 261050032                 | 261050070                 | 261050093                 | 261180018                 |
| 261050035                 | 261050071                 | 261050402                 | 261180022                 |
| 261050042                 | 261050072                 | 261050406                 | 261290014                 |
| 261050045                 | 261050074                 | 261050601                 | 261290016                 |
| 261050047                 | 261050076                 | 261050605                 |                           |

**Problem 1****[40 marks]***Steady 2D conduction in a rectangular plate (Cartesian coordinates)*

A thin rectangular plate of width  $a$  and height  $b$  has three sides held at  $0^\circ\text{C}$  while the fourth (top) side is held at a uniform temperature  $T_0$  (Fig. 1). There is no internal heat generation and the plate is thin enough that conduction may be treated as two-dimensional, so that steady-state temperature field satisfies Laplace's equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0, \quad 0 \leq x \leq a, \quad 0 \leq y \leq b.$$

**Boundary conditions:**

$$T(x, 0) = 0, \quad T(0, y) = 0, \quad T(a, y) = 0, \quad T(x, b) = T_0.$$

**Data:**  $a = 0.4$  m,  $b = 0.2$  m,  $T_0 = 100^\circ\text{C}$ .**Analytical solution** (by separation of variables):

$$T(x, y) = \frac{4T_0}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi x}{a}\right) \frac{\sinh(n\pi y/a)}{\sinh(n\pi b/a)}.$$

Use enough terms to ensure the series has converged to at least five significant figures before using it as your benchmark.

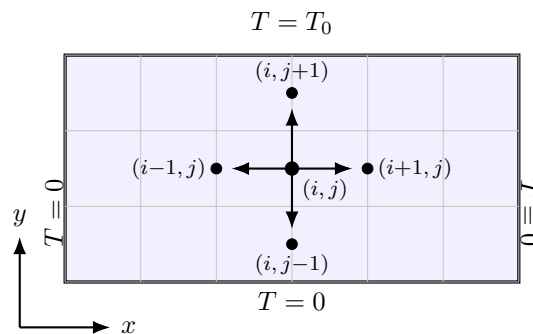


Fig. 1 — Rectangular plate with three sides at  $0^\circ\text{C}$  and the top side at  $T_0$ ; inset shows the interior 5-point stencil at node  $(i, j)$

**Tasks:**

- Derive the standard 5-point finite-difference equation at an interior node, and assemble the global linear system  $[A]\{T\} = \{b\}$  for a uniform grid with  $\Delta x = \Delta y = h$ .
- Solve the system using a *direct* method (Gaussian elimination or LU decomposition). Record the CPU time and memory used as  $h$  is refined ( $h = 0.02, 0.01, 0.005$  m).
- Solve the same system using Jacobi, Gauss-Seidel, and SOR (try at least three values of the relaxation factor  $\omega$ , including the theoretical optimum for a rectangular Dirichlet problem). Record the number of iterations and CPU time for each.
- Compare all solutions with the analytical series at the points  $(a/4, b/2)$ ,  $(a/2, b/2)$ ,  $(3a/4, b/2)$ , and  $(a/2, 3b/4)$ . Tabulate the maximum absolute error versus  $h$  and confirm the expected  $O(h^2)$  accuracy of the 5-point scheme.

- (e) Summarize, in a short paragraph, how solution time scales with the number of unknowns  $N$  for the direct method versus each iterative method, and recommend a solver for this problem class.

## Problem 2

[40 marks]

*Steady 2-D Conduction with Uniform Internal Heat Generation in a Rectangular Bar*

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A long, electrically heated bar has a rectangular cross-section of dimensions  $2a \times 2b$  and carries a uniform internal heat generation rate  $q'''$  (W/m<sup>3</sup>) due to resistive heating. The bar has thermal conductivity  $k$ , and all four outer surfaces exchange heat by convection with a fluid at temperature  $T_\infty$  and heat transfer coefficient  $h$ . By symmetry, only one quadrant of the cross-section needs to be analyzed.

Governing equation (quadrant domain,  $0 \leq x \leq a$ ,  $0 \leq y \leq b$ ):

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{q'''}{k} = 0$$

Boundary conditions:

$$x = 0 : \frac{\partial T}{\partial x} = 0 \quad (\text{symmetry, insulated})$$

$$y = 0 : \frac{\partial T}{\partial y} = 0 \quad (\text{symmetry, insulated})$$

$$x = a : -k \frac{\partial T}{\partial x} = h(T - T_\infty)$$

$$y = b : -k \frac{\partial T}{\partial y} = h(T - T_\infty)$$

Data  $2a = 0.10$  m,  $2b = 0.06$  m,  $k = 15$  W/m · K,  $q''' = 2 \times 10^6$  W/m<sup>3</sup>,  $h = 500$  W/m<sup>2</sup> · K,  $T_\infty = 30^\circ\text{C}$ .

Task: Using a finite-difference formulation with a uniform grid  $\Delta x = \Delta y$ :

- Derive nodal equations for interior nodes, nodes on the two insulated boundaries, nodes on the two convective boundaries, and the two corner nodes (insulated–insulated and insulated–convective).
- Solve for the steady-state temperature distribution (e.g., by Gauss–Seidel or SOR).
- Report the maximum temperature (occurring at  $x = y = 0$ ) and plot isotherms over the full cross-section.
- Verify an overall energy balance: total heat generated per unit length should equal total heat lost by convection per unit length.

## Problem 3

[40 marks]

*Steady 2D conduction in a solid cylinder (cylindrical/polar coordinates)*

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A long solid cylinder of radius  $R$  has its curved outer surface held at a temperature that varies with the circumferential angle  $\theta$  (Fig. 2). There is no axial ( $z$ ) variation and no internal heat generation, so the temperature field  $T(r, \theta)$  satisfies

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} = 0, \quad 0 \leq r \leq R, \quad 0 \leq \theta < 2\pi,$$

subject to boundedness of  $T$  at  $r = 0$  and periodicity,  $T(r, \theta) = T(r, \theta + 2\pi)$ .

**Boundary condition:**

$$T(R, \theta) = T_0 + \Delta T_1 \cos \theta + \Delta T_2 \cos 2\theta.$$

**Data:**  $R = 0.08$  m,  $T_0 = 80^\circ\text{C}$ ,  $\Delta T_1 = 25^\circ\text{C}$ ,  $\Delta T_2 = 10^\circ\text{C}$ ,  $k = 15$  W/(m K).

**Analytical solution** (separation of variables; the  $r^{-n}$  and  $\ln r$  terms are discarded because  $T$  must remain bounded at  $r = 0$ ):

$$T(r, \theta) = T_0 + \Delta T_1 \left( \frac{r}{R} \right) \cos \theta + \Delta T_2 \left( \frac{r}{R} \right)^2 \cos 2\theta.$$

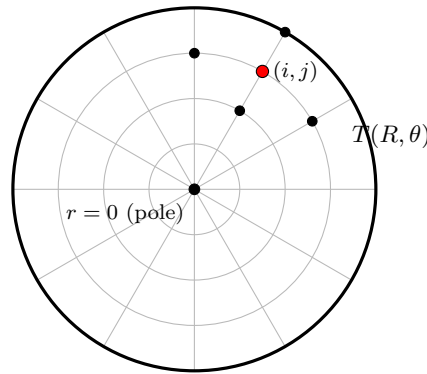


Fig. 2 — Polar grid for the solid cylinder: concentric circles ( $r = \text{const.}$ ) and radial spokes ( $\theta = \text{const.}$ ); the pole at  $r = 0$  requires special treatment

**Tasks:**

- Derive the finite-difference equation at a generic interior node  $(i, j)$  using a uniform grid  $\Delta r = R/M$ ,  $\Delta \theta = 2\pi/N$ , and impose periodicity ( $j = N + 1 \equiv j = 1$ ) in  $\theta$ .
- At the pole ( $r = 0$ ), show, using an energy balance on a small disk of radius  $\Delta r/2$ , that the finite-difference equation reduces to

$$T_0 = \frac{1}{N} \sum_{j=1}^N T_{1,j},$$

i.e., the centre temperature equals the average of all nodes on the first ring. Use this as your pole boundary condition.

- Assemble and solve  $[A]\{T\} = \{b\}$  by (i) a direct method and (ii) Jacobi, Gauss–Seidel, and SOR. Comment on how the periodic boundary in  $\theta$  changes the bandwidth/sparsity pattern of  $[A]$  compared with Problem 1, and how this affects the relative efficiency of direct versus iterative solvers.
- Compare your numerical  $T(r, \theta)$  with the analytical solution at  $r = R/4, R/2, 3R/4, R$  for  $\theta = 0^\circ, 45^\circ, 90^\circ, 180^\circ$ . Report the maximum error and its grid dependence for  $M, N$  each doubled twice.

- (e) Compute the surface heat flux  $q''(\theta) = -k \partial T / \partial r|_{r=R}$  from your numerical solution and verify numerically that  $\oint_0^{2\pi} q''(\theta) R d\theta \approx 0$  (a required energy-balance check, since there is no internal generation).

## Problem 4

[40 marks]

*Conduction with temperature-dependent thermal conductivity and internal heat generation*

A plane wall of thickness  $L$  has a temperature-dependent thermal conductivity

$$k(T) = k_0 (1 + \beta T)$$

and generates heat uniformly at a volumetric rate  $q'''$ . The wall is thin enough in the other two directions that conduction is one-dimensional and steady, so

$$\frac{\partial}{\partial x} \left[ k(T) \frac{\partial T}{\partial x} \right] + q''' = 0, \quad 0 \leq x \leq L,$$

with  $T(0) = T_1$  and  $T(L) = T_2$  (Fig. 3).

**Data:**  $L = 0.1$  m,  $k_0 = 15$  W/(mK),  $\beta = 0.002$  K<sup>-1</sup>,  $q''' = 2 \times 10^5$  W/m<sup>3</sup>,  $T_1 = 250^\circ\text{C}$ ,  $T_2 = 50^\circ\text{C}$ .

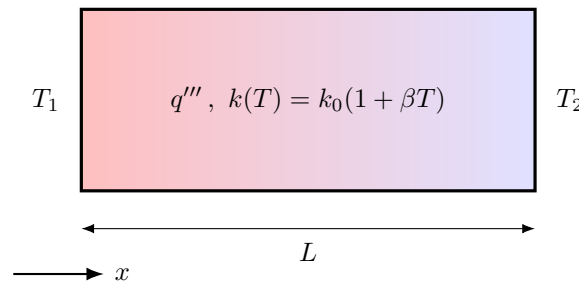


Fig. 3 — Plane wall with temperature-dependent conductivity and uniform internal heat generation

**Analytical (benchmark) solution.** Introduce the Kirchhoff transform

$$\Theta(T) = \int_0^T \frac{k(T')}{k_0} dT' = T + \frac{\beta}{2} T^2,$$

so that  $k_0 d\Theta/dx = k(T) dT/dx$  and the governing equation becomes the linear equation  $d^2\Theta/dx^2 = -q'''/k_0$ . Integrating twice and applying the boundary conditions ( $\Theta_1 = T_1 + \frac{\beta}{2} T_1^2$ ,  $\Theta_2 = T_2 + \frac{\beta}{2} T_2^2$ ) gives

$$\Theta(x) = \Theta_1 + (\Theta_2 - \Theta_1) \frac{x}{L} + \frac{q'''}{2k_0} x(L - x),$$

and inverting  $\Theta = T + \frac{\beta}{2} T^2$  (physical root) gives the exact temperature distribution

$$T(x) = \frac{-1 + \sqrt{1 + 2\beta \Theta(x)}}{\beta}.$$

As a check, note that setting  $\beta = 0$  recovers the familiar constant-property result  $T(x) = T_1 + (T_2 - T_1) \frac{x}{L} + \frac{q'''}{2k_0} x(L - x)$ .

**Tasks:**

- Derive the above Kirchhoff-transform solution yourself (show all steps), and evaluate  $T(x)$  at  $x = 0, L/4, L/2, 3L/4, L$ .
- Discretize the original nonlinear equation directly (without the Kirchhoff transform) using finite differences with  $k$  evaluated at the face between nodes (e.g.  $k_{i+1/2} = k_0[1 + \beta(T_i + T_{i+1})/2]$ ). Because the coefficients depend on the unknown  $T$ , use a Picard (successive-substitution) iteration: guess a  $T$  field, update all  $k_{i\pm 1/2}$ , solve the resulting *linear* tridiagonal system (Thomas algorithm), and repeat until  $\max |T^{p+1} - T^p| < 10^{-6}$ .
- Verify your code first for  $\beta = 0$  against the constant-property formula above, then run it for  $\beta = 0.002 \text{ K}^{-1}$  and compare with the exact Kirchhoff-transform solution at the same five points. Report the number of Picard iterations required.
- Repeat for  $\beta = 0.005$  and  $\beta = -0.001 \text{ K}^{-1}$  (conductivity increasing vs. decreasing with temperature). Discuss how  $\beta$  changes the curvature of the temperature profile and the number of iterations needed for convergence.
- (Optional) Compare the Picard iteration above with a Newton linearization of the nonlinear finite-difference equations, in terms of the number of iterations required for the same convergence tolerance.

## Problem 5

[60 marks]

### *Solid cylinder under asymmetric circumferential heat flux*

A long solid cylinder of radius  $R$  and thermal conductivity  $k$  loses heat by convection ( $h, T_\infty$ ) over its *entire* outer surface while simultaneously receiving a prescribed, non-uniform (asymmetric) heat flux  $q''(\theta)$  over that same surface, representing an external radiant source that is stronger on one side of the cylinder than the other (Fig. 4). There is no internal heat generation. The steady-state surface energy balance is

$$-k \left. \frac{\partial T}{\partial r} \right|_{r=R} = h(T(R, \theta) - T_\infty) - q''(\theta), \quad 0 \leq \theta < 2\pi,$$

with  $T$  bounded at  $r = 0$  and periodic in  $\theta$ .

**Prescribed flux (fully asymmetric — not symmetric about any axis):**

$$q''(\theta) = q_0 + q_1 \cos \theta + q_2 \sin \theta + q_3 \cos 3\theta.$$

**Data:**  $R = 0.05 \text{ m}$ ,  $k = 40 \text{ W/(mK)}$ ,  $h = 25 \text{ W/(m}^2\text{K)}$ ,  $T_\infty = 25^\circ\text{C}$ ,  $q_0 = 2000$ ,  $q_1 = 1200$ ,  $q_2 = 800$ ,  $q_3 = 500$  (all  $\text{W/m}^2$ ).

**Analytical solution.** Writing  $T(r, \theta) - T_\infty = A_0 + \sum_{n \geq 1} (r/R)^n (A_n \cos n\theta + B_n \sin n\theta)$  and substituting into the surface balance term by term gives, for each Fourier mode  $n$  present in  $q''(\theta)$ ,

$$A_0 = \frac{q_0}{h}, \quad A_n = \frac{q_n}{h + nk/R}, \quad B_n = \frac{q_n^{(s)}}{h + nk/R},$$

where  $q_n, q_n^{(s)}$  are the cosine and sine coefficients of  $q''(\theta)$  at harmonic  $n$ . For the data above,

$$T(r, \theta) = 105 + 1.4545 \left(\frac{r}{R}\right) \cos \theta + 0.9697 \left(\frac{r}{R}\right) \sin \theta + 0.2062 \left(\frac{r}{R}\right)^3 \cos 3\theta \quad [^\circ\text{C}].$$

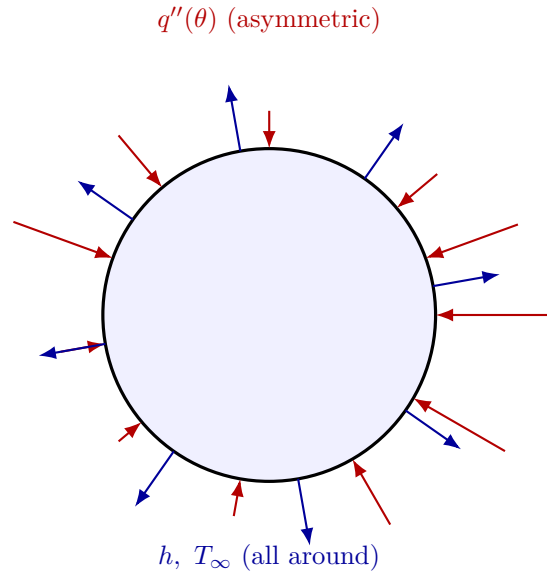


Fig. 4 — Cylinder with a fully asymmetric external flux  $q''(\theta)$  (red, length  $\propto$  local magnitude) and convective cooling over the whole surface (blue)

### Tasks:

- Derive the discretized Robin boundary condition at  $r = R$  (a ghost-node or one-sided-difference formulation is acceptable), and solve for  $T(r, \theta)$  on a polar grid (reuse your pole treatment from Problem 2).
- Verify your assembled Fourier coefficients  $A_0, A_1, B_1, A_3$  against the numeric values implied above before running your solver.
- Compare your finite-difference  $T(R, \theta)$  with the analytical formula at  $\theta = 0^\circ, 60^\circ, 120^\circ, 180^\circ, 270^\circ$ , and report the maximum error.
- Locate the angular position of the hottest and coldest points on the surface (analytically, by differentiating  $T(R, \theta)$  with respect to  $\theta$ ; and numerically, from your computed surface temperatures) and compare.
- Confirm energy conservation: the net heat flow across the surface,

$$\oint_0^{2\pi} [q''(\theta) - h(T(R, \theta) - T_\infty)] R d\theta,$$

should numerically integrate to (approximately) zero.

## Problem 6

[60 marks]

### Composite (two-layer) cylindrical shell

An insulated pipe consists of two concentric layers of different material: an inner metal wall (thermal conductivity  $k_1$ ) from  $r_0$  to  $r_1$ , and an outer insulation layer (thermal conductivity  $k_2$ ) from  $r_1$  to  $r_2$  (Fig. 5). The bore surface at  $r_0$  is held at a fixed temperature  $T_i$  (e.g., by a hot fluid), and the outer surface at  $r_2$  loses heat by convection to ambient air at  $T_\infty$  with coefficient  $h$ . There is no internal heat generation, so in each layer

$$\frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) = 0.$$

**Interface conditions at  $r = r_1$ :** continuity of temperature and of radial heat flux,

$$T_1(r_1) = T_2(r_1), \quad k_1 \frac{\partial T_1}{\partial r} \Big|_{r_1} = k_2 \frac{\partial T_2}{\partial r} \Big|_{r_1}.$$

**Data:**  $r_0 = 0.02$  m,  $r_1 = 0.03$  m,  $r_2 = 0.08$  m,  $k_1 = 45$  W/(m K) (steel),  $k_2 = 0.05$  W/(m K) (insulation),  $T_i = 150^\circ\text{C}$ ,  $T_\infty = 25^\circ\text{C}$ ,  $h = 15$  W/(m<sup>2</sup> K).

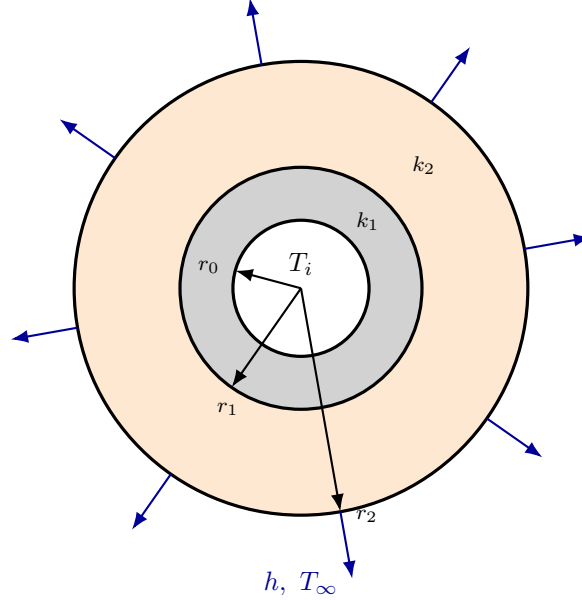


Fig. 5 — Composite cylindrical shell: metal wall ( $k_1$ ,  $r_0 \rightarrow r_1$ ) and insulation ( $k_2$ ,  $r_1 \rightarrow r_2$ ), with convective outer boundary

**Analytical (thermal-resistance) solution.** Per unit length,

$$R'_1 = \frac{\ln(r_1/r_0)}{2\pi k_1}, \quad R'_2 = \frac{\ln(r_2/r_1)}{2\pi k_2}, \quad R'_3 = \frac{1}{2\pi r_2 h}, \quad Q' = \frac{T_i - T_\infty}{R'_1 + R'_2 + R'_3},$$

and the temperature in each layer follows the logarithmic profile

$$T(r) = T_i - \frac{Q'}{2\pi k_1} \ln\left(\frac{r}{r_0}\right) \quad (r_0 \leq r \leq r_1), \quad T(r) = T(r_1) - \frac{Q'}{2\pi k_2} \ln\left(\frac{r}{r_1}\right) \quad (r_1 \leq r \leq r_2).$$

### Tasks:

- Compute  $R'_1$ ,  $R'_2$ ,  $R'_3$ ,  $Q'$ ,  $T(r_1)$ , and  $T(r_2)$  from the formulas above, and verify  $T(r_2)$  is consistent with the convective boundary condition.
- Discretize the domain with the interface exactly on a grid point  $r = r_1$ . For the interior nodes in each layer use the standard cylindrical finite-difference equation; at the interface node, derive the modified equation using a control-volume energy balance in which each half-control-volume uses its own material's conductivity (do *not* simply average  $k_1$  and  $k_2$  arithmetically).
- Solve the resulting tridiagonal system directly (Thomas algorithm) and also by Gauss–Seidel iteration. Compare both to the analytical  $T(r)$  above at  $r = r_0, r_1, (r_1 + r_2)/2, r_2$ , and report the maximum error for grid spacings  $\Delta r = 2.5, 1.0, 0.5$  mm.

- (d) Since this problem is one-dimensional, explain — referring back to your findings in Problems 1 and 2 — why a direct (Thomas algorithm) solve is essentially always preferable here, whereas for the 2D problems the choice between direct and iterative solvers was less clear-cut.
- (e) (*Optional/bonus*) Suppose the two layers are not in perfect contact, so that an interfacial (contact) resistance  $R_c''$  exists at  $r_1$ , giving a temperature jump  $T_1(r_1) - T_2(r_1) = Q' R_c'' / (2\pi r_1)$ . Modify both the analytical solution and your interface finite-difference equation to include  $R_c'' = 5 \times 10^{-4} \text{ m}^2 \text{ K/W}$ , and quantify its effect on  $T(r_2)$  and  $Q'$ .

## Problem 7

[60 marks]

### *Eccentric circular duct in a square refractory block*

Problem 4.9 considered a circular duct of radius  $R$ , running through a square refractory block of side  $W$ , *centered* on the block's axis (Fig. 6a). Here we keep the same physical situation but move the duct off-centre by an eccentricity  $e$  (Fig. 6b), and additionally ask you to benchmark your irregular-boundary treatment against a known analytical shape factor before trusting it on the eccentric case.

**Data:**  $W = 1.0 \text{ m}$ ,  $D = 2R = 0.2 \text{ m}$ ,  $k = 1.04 \text{ W/(m K)}$ , inner (duct) surface  $T_i = 1200^\circ\text{C}$ , outer (block) surface  $T_o = 50^\circ\text{C}$ . Eccentricity for Part (c) onward:  $e = 0.15 \text{ m}$  (duct centre shifted along the horizontal axis only, so the duct remains fully inside the block).

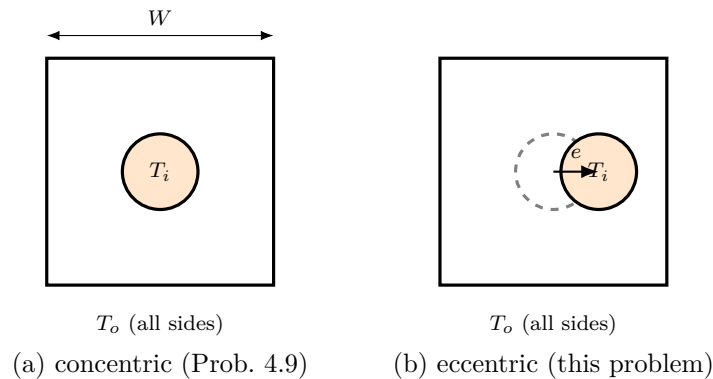


Fig. 6 — (a) Concentric duct as in Problem 4.9; (b) duct shifted by eccentricity  $e$  along the horizontal axis (dashed circle shows the original concentric position)

**The irregular-boundary (“irregular star”) finite-difference formula.** At a grid node  $O$  adjacent to the curved duct surface, the neighbouring points to the west, east, south, and north lie at fractional distances  $\alpha_w \Delta x$ ,  $\alpha_e \Delta x$ ,  $\alpha_s \Delta y$ ,  $\alpha_n \Delta y$  from  $O$  (each  $\alpha = 1$  for a normal interior neighbour, or  $< 1$  where the curved boundary cuts the grid line before reaching the next regular node, in which case the corresponding temperature is the known boundary value  $T_i$ ). The consistent finite-difference form of Laplace’s equation at such a node is

$$\frac{2}{\Delta x^2} \left[ \frac{T_W}{\alpha_w(\alpha_w + \alpha_e)} + \frac{T_E}{\alpha_e(\alpha_w + \alpha_e)} - \frac{T_O}{\alpha_w \alpha_e} \right] + \frac{2}{\Delta y^2} \left[ \frac{T_S}{\alpha_s(\alpha_s + \alpha_n)} + \frac{T_N}{\alpha_n(\alpha_s + \alpha_n)} - \frac{T_O}{\alpha_s \alpha_n} \right] = 0,$$

which reduces to the familiar 5-point formula when all four  $\alpha = 1$ .

**Analytical benchmark (concentric case only).** For a circular duct of diameter  $D$  centred in an infinite (or thick,  $W > 1.4D$ ) square solid of side  $W$  with isothermal surfaces, the conduction

shape factor is the standard result

$$\frac{S}{L} = \frac{2\pi}{\ln(1.08 W/D)}, \quad \frac{Q}{L} = \frac{S}{L} k (T_i - T_o).$$

No comparably simple closed form exists once the duct is eccentric — this is precisely why Part (c) is done numerically.

### Tasks:

- For the **concentric** case, use the symmetry of the geometry to model only one-eighth (or one-quarter, if simpler) of the domain. Assemble the finite-difference equations (using the irregular-star formula above for nodes next to the duct) and solve for  $T(x, y)$  by (i) a direct method and (ii) Gauss–Seidel/SOR.
- Compute  $Q/L$  from your numerical solution (by summing the flux either around the duct or into the outer boundary) and compare it with the shape-factor result above. Perform a grid-refinement study ( $\Delta x = \Delta y = 5, 2.5, 1.25$  cm) and report how the error in  $Q/L$  decreases with grid size — note whether you still observe clean second-order convergence, given that the irregular-star formula is used near the curved boundary.
- For the **eccentric** case ( $e = 0.15$  m), identify which symmetry (if any) survives the offset, and solve on the correspondingly reduced domain. Report  $Q/L$  and compare it with the concentric shape-factor value; quantify the percentage change in  $Q/L$  caused by the eccentricity.
- Repeat the direct-vs-iterative solver comparison (iteration count, CPU time) for the eccentric case, and comment on whether the loss of full symmetry changes the number of iterations needed for Gauss–Seidel/SOR to converge, relative to the concentric case of the same grid size.
- Plot isotherms for both the concentric and eccentric cases and discuss, physically, why the duct wall closest to the nearby block face runs cooler (or hotter) relative to the concentric case.

## Problem 8

[60 marks]

*Hollow cylinder with an insulated bore under one-sided flux*

Problem 4.11 considered a *solid* cylinder, exposed on one half of its circumference to a solar-type flux and cooled by convection over its whole periphery. Here the same heating and cooling arrangement is applied to a **hollow** cylinder (a tube) of inner radius  $R_i$  and outer radius  $R_o$ , whose inner bore is insulated (Fig. 7) — for example, a tube whose bore carries no flow, or an electrical/solar collector conduit backed by insulation.

**Governing equation** (steady, no generation):

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} = 0, \quad R_i \leq r \leq R_o, \quad 0 \leq \theta < 2\pi.$$

**Boundary conditions:**

$$\left. \frac{\partial T}{\partial r} \right|_{r=R_i} = 0 \quad (\text{insulated bore}), \quad -k \left. \frac{\partial T}{\partial r} \right|_{r=R_o} = h(T(R_o, \theta) - T_\infty) - q''(\theta),$$

$$q''(\theta) = \begin{cases} q_0'' \sin \theta, & 0 < \theta < \pi \\ 0, & \pi < \theta < 2\pi \end{cases} \quad (\text{identical flux definition to Problem 4.11}).$$

**Data:**  $R_i = 0.02$  m,  $R_o = 0.05$  m,  $k = 20$  W/(m K),  $h = 15$  W/(m<sup>2</sup> K),  $T_\infty = 25^\circ\text{C}$ ,  $q_0'' = 1200$  W/m<sup>2</sup>.

**Analytical solution.** Because the bore at  $r = R_i$  is insulated (rather than the origin being part of the domain as in Problem 4.11), the general bounded-at- $r = 0$  solution no longer applies; instead each harmonic must independently satisfy the zero-gradient condition at  $R_i$ . Define

$$f_n(r) = r^n + R_i^{2n} r^{-n},$$

which automatically satisfies  $f_n'(R_i) = 0$  for every  $n$ . Writing  $q''(\theta) = a_0 + \sum_{n \geq 1} (a_n \cos n\theta + b_n \sin n\theta)$  (the same half-wave-rectified sine expansion used in Problem 4.11:  $a_0 = q_0''/\pi$ ,  $b_1 = q_0''/2$ ,  $a_{2m} = -2q_0''/[\pi(4m^2 - 1)]$  for  $m = 1, 2, 3, \dots$ , and all other coefficients zero), the general solution satisfying both boundary conditions is

$$T(r, \theta) = T_\infty + \frac{a_0}{h} + \frac{b_1 f_1(r)}{h f_1(R_o) + k f_1'(R_o)} \sin \theta + \sum_{m=1}^{\infty} \frac{a_{2m} f_{2m}(r)}{h f_{2m}(R_o) + k f_{2m}'(R_o)} \cos 2m\theta.$$

**Check:** as  $R_i \rightarrow 0$ ,  $f_n(r) \rightarrow r^n$ ,  $f_n(R_o) \rightarrow R_o^n$ ,  $f_n'(R_o) \rightarrow n R_o^{n-1}$ , and the expression above reduces exactly to the Problem 4.11 solid-cylinder result.

#### Tasks:

- Derive the boundary-condition matching that produces  $f_n(r)$  and the coefficient formula above (i.e., show the algebra that leads from the general  $r^n/r^{-n}$  solution to the stated result).
- Discretize the annulus on a grid  $\Delta r = (R_o - R_i)/M$ ,  $\Delta \theta = 2\pi/N$  (periodic in  $\theta$ ). Unlike Problems 2 and 4 above, no pole treatment is required here — instead derive the discretized insulated-bore condition at  $r = R_i$  (ghost-node or one-sided-difference formulation).
- Solve  $[A]\{T\} = \{b\}$  by a direct method and by Gauss–Seidel/SOR. Compare your numerical  $T(R_o, \theta)$  with the analytical series at  $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ$ , and locate the hottest and coldest points on the outer surface.
- Verify energy conservation:  $\oint_0^{2\pi} [q''(\theta) - h(T(R_o, \theta) - T_\infty)] R_o d\theta \approx 0$ , and confirm separately that the heat flux at  $r = R_i$  is numerically zero everywhere (checking your insulated-bore implementation).
- Limiting-case validation:** re-run your annulus code with  $R_i$  reduced to  $0.001 R_o$  (nearly solid) and confirm that your computed  $T(r, \theta)$  converges to the original Problem 4.11 solid-cylinder analytical solution (using the same  $R_o, k, h, T_\infty, q_0''$  as Problem 4.11's data). Report the residual difference as  $R_i \rightarrow 0$ .